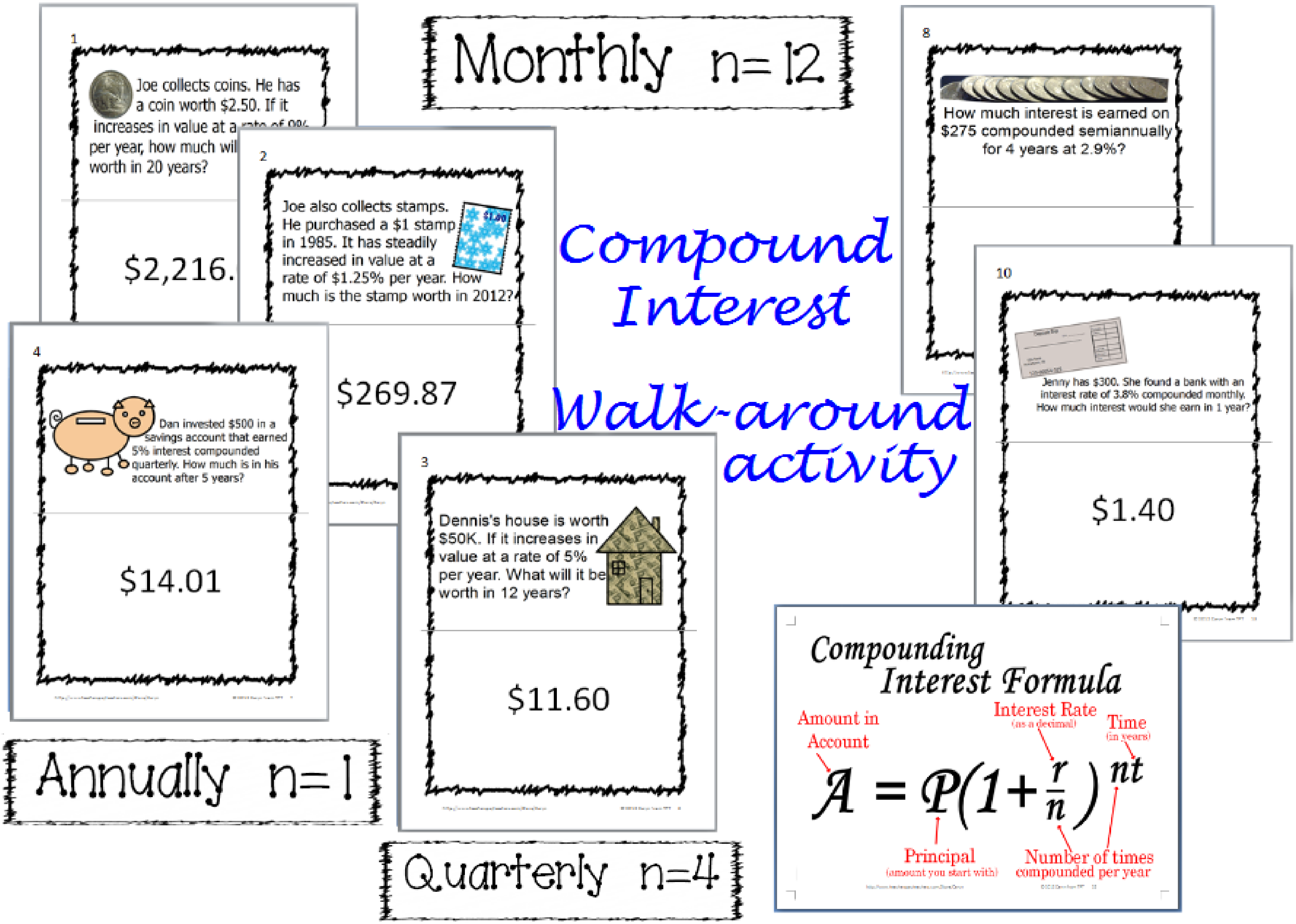

Compound Interest
Walk-around activity
Monthly n=12
Annually n=1
Quarterly n=4
1
Joe collects coins. He has a coin worth $2.50. If it increases in value at a
per year, how much wil
worth in 20 years?
$2,216.
2
Joe also collects stamps. He purchased a $1 stamp in 1985. It has steadily increased in value at a rate of $1.25% per year. How much is the stamp worth in 2012?
$269.87
4
Dan invested $500 in a savings account that earned 5% interest compounded quarterly. How much is in his account after 5 years?
$14.01
3
Dennis's house is worth $50K. If it increases in value at a rate of 5% per year. What will it be worth in 12 years?
$11.60
8
How much interest is earned on $275 compounded semiannually for 4 years at 2.9%?
10
Jenny has $300. She found a bank with an interest rate of 3.8% compounded monthly. How much interest would she earn in 1 year?
$1.40
Compounding Interest Formula
$A = P(1+\frac{r}{n})^{nt}$
Amount in Account
Interest Rate (as a decimal)
Time (in years)
Principal (amount you start with)
Number of times compounded per year

Table of Contents

Instructions

Directions for instructor

This activity is designed to help students practice compound interest type problems. This activity also gets students up and about. Place the 10 cards on the wall around your room. Students pick any card to begin with. They should use the compound interest formula to solve the problem. After working the problem on their own paper, students look around the room for the solution that matches theirs. Then they work the problem on this page and look for the new solution. They should continue until they return to the card they started with.

Included are 2 version of this activity.

- One with 10 cards for use as a walk-around activity A shorter version with only 5 cards great for table work, bell work or use as a review.

To help with grading, I have included 2 student answer sheets: One that includes with room to show work, and another with just boxes.

Possible Uses

- Mid-Lesson or End of Lesson Check for understanding
- Math Station for students that have finished work early
- Test Review
- Homework Alternative
- Bell work

Hints and suggestions:

When making copies of the shorter version, I use different colored paper. This allows for easier identification when students misplace a piece.

I have included a sheet with all 10 problems on it. This works great for absent students or for students with difficulties walking around the room (broken legs, wheel chairs, autism or other special condition). I also give this to our special ed paras, this way they can help student during their study skills classes.

Have Fun. Thanks for your purchase.

1

Joe collects coins. He has a coin worth $2.50. If it increases in value at a rate of 9% per year, how much will it be worth in 20 years?

$2,216.02

2

Joe also collects stamps. He purchased a $1 stamp in 1985. It has steadily increased in value at a rate of $1.25% per year. How much is the stamp worth in 2012?

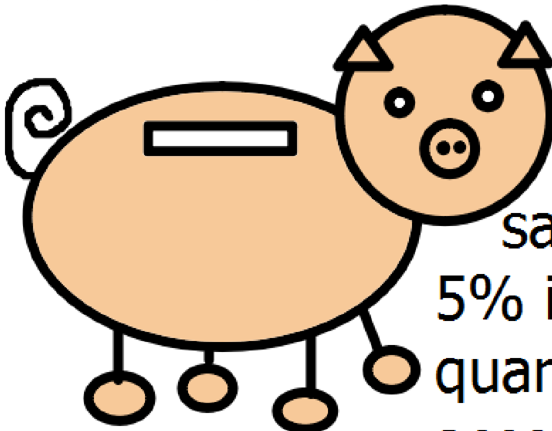
$269.87

3

Dennis's house is worth $50K. If it increases in value at a rate of 5% per year. What will it be worth in 12 years?

$11.60

4



Dan invested $500 in a savings account that earned 5% interest compounded quarterly. How much is in his account after 5 years?

$14.01

5

Pauly placed $500 in an IRA when he was 20. He predicts that it will increase in value by 6% per year. The IRA compounds quarterly. How much does he expect his IRA to be worth in 25 years?

$13,024.74

6

Jim invested $250 when he was 18. His investment earned 8.5% interest compounded quarterly. How much would his investment be worth when he is 65?

$239,898

7

Katy deposited $200 in her account. If her bank offers 3% interest compounded monthly, how much will be in her account in 10 years?

$641.02

8

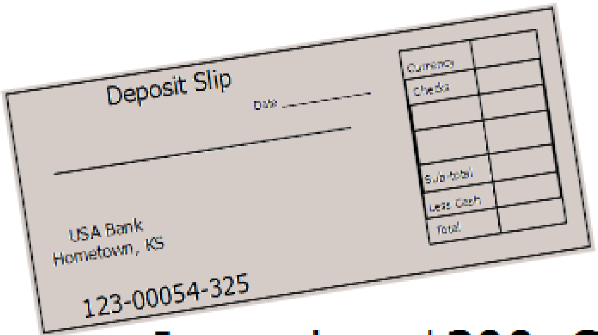
How much interest is earned on $275 compounded semiannually for 4 years at 2.9%?

$89,793

Sally purchased a condo for $120,000 in 2001. If the condo increased in value by 6.5% each year, how much is the condo worth in 2012?

$33.57

10



Jenny has $300. She found a bank with an interest rate of 3.8% compounded monthly. How much interest would she earn in 1 year?

$1.40

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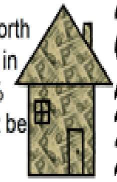
Joe also collects stamps. He purchased a $1 stamp in 1985. It has steadily increased in value at a rate of $1.25% per year. How much is the stamp worth in 2012?

$269.87

3



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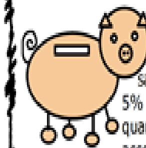
4



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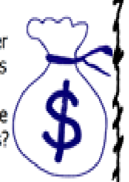
7



Katy deposited $200 in her account. If her bank offers 3% interest compounded monthly, how much will be in her account in 10 years?

$641.02

8

How much interest is earned on $275 compounded semiannually for 4 years at 2.9%?

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9

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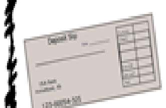


Jenny has $300. She found a bank with an interest rate of 3.8% compounded monthly. How much interest would she earn in 1 year?

$1.40

Student Answer Sheet – Version A1

Student Answer Sheet

Name ____________________

Period _____

Pick a card to start with. Write the number in the 1st box. Work the problem and then locate the card with the correct answer. Write the card number in the 2nd box, and then continue until you reach your first card again.

Compound Interest Formula

$\mathcal{A} =$

Show work below

____ $\mathcal{P}$ = r = n = t = $\mathcal{A}$ =	____ $\mathcal{P}$ = r = n = t = $\mathcal{A}$ =	____ $\mathcal{P}$ = r = n = t = $\mathcal{A}$ =	____ $\mathcal{P}$ = r = n = t = $\mathcal{A}$ =
____ $\mathcal{P}$ = r = n = t = $\mathcal{A}$ =	____ $\mathcal{P}$ = r = n = t = $\mathcal{A}$ =	____ $\mathcal{P}$ = r = n = t = $\mathcal{A}$ =	____ $\mathcal{P}$ = r = n = t = $\mathcal{A}$ =
____ $\mathcal{P}$ = r = n = t = $\mathcal{A}$ =	____ $\mathcal{P}$ = r = n = t = $\mathcal{A}$ =		

Student Answer Sheet – Version A2

Student Answer Sheet

Name ______________________

Period _____

Pick a card to start with. Write the number in the 1st box. Work the problem and then locate the card with the correct answer. Write the card number in the 2nd box, and then continue until you reach your first card again.

Student Answer Sheet

Name ______________________

Period _____

Pick a card to start with. Write the number in the 1st box. Work the problem and then locate the card with the correct answer. Write the card number in the 2nd box, and then continue until you reach your first card again.

Student Answer Sheet

Name ______________________

Period _____

Pick a card to start with. Write the number in the 1st box. Work the problem and then locate the card with the correct answer. Write the card number in the 2nd box, and then continue until you reach your first card again.

Student Answer Sheet

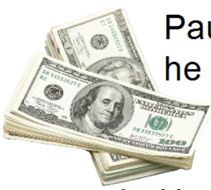
Name ______________________

Period _____

Pick a card to start with. Write the number in the 1st box. Work the problem and then locate the card with the correct answer. Write the card number in the 2nd box, and then continue until you reach your first card again.

Version B

1



Pauly placed $500 in an IRA when he was 20. He predicts that it will increase in value by 6% per year. The IRA compounds quarterly. How much does he expect his IRA to be worth in 25 years?

$11.60

2

Jim invested $250 when he was 18. His investment earned 8.5% interest compounded quarterly. How much would his investment be worth when he is 65?

$14.01

3

Joe collects coins. He has a coin worth $2.50. If it increases in value at a rate of 9% per year, how much will it be worth in 20 years?

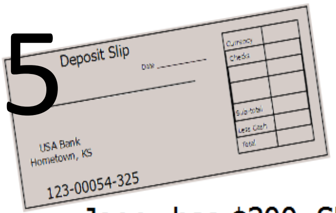
$269.87

4

Katy deposited $200 in her account. If her bank offers 3% interest compounded monthly, how much will be in her account in 10 years?

$2216.02

5



Jenny has $300. She found a bank with an interest rate of 3.8% compounded monthly. How much interest would she earn in 1 year?

$13,024.74

Student Answer Sheet – Version B1

Student Answer Sheet

Name ____________________

Period _____

Pick a card to start with. Write the number in the 1st box. Work the problem and then locate the card with the correct answer. Write the card number in the 2nd box, and then continue until you reach your first card again.

Compound Interest Formula

$\mathcal{A} =$

Show work below

____	____	____	____	____
$\mathcal{P} =$	$\mathcal{P} =$	$\mathcal{P} =$	$\mathcal{P} =$	$\mathcal{P} =$
$r =$	$r =$	$r =$	$r =$	$r =$
$n =$	$n =$	$n =$	$n =$	$n =$
$t =$	$t =$	$t =$	$t =$	$t =$
$\mathcal{A} =$	$\mathcal{A} =$	$\mathcal{A} =$	$\mathcal{A} =$	$\mathcal{A} =$

Student Answer Sheet

Name ____________________

Period _____

Pick a card to start with. Write the number in the 1st box. Work the problem and then locate the card with the correct answer. Write the card number in the 2nd box, and then continue until you reach your first card again.

Compound Interest Formula

$\mathcal{A} =$

Show work below

____	____	____	____	____
$\mathcal{P} =$	$\mathcal{P} =$	$\mathcal{P} =$	$\mathcal{P} =$	$\mathcal{P} =$
$r =$	$r =$	$r =$	$r =$	$r =$
$n =$	$n =$	$n =$	$n =$	$n =$
$t =$	$t =$	$t =$	$t =$	$t =$
$\mathcal{A} =$	$\mathcal{A} =$	$\mathcal{A} =$	$\mathcal{A} =$	$\mathcal{A} =$

Student Answer Sheet – Version B2

Student Answer Sheet

Name ____________________

Period _____

Pick a card to start with. Write the number in the 1st box. Work the problem and then locate the card with the correct answer. Write the card number in the 2nd box, and then continue until you reach your first card again.

Student Answer Sheet

Name ____________________

Period _____

Pick a card to start with. Write the number in the 1st box. Work the problem and then locate the card with the correct answer. Write the card number in the 2nd box, and then continue until you reach your first card again.

Student Answer Sheet

Name ____________________

Period _____

Pick a card to start with. Write the number in the 1st box. Work the problem and then locate the card with the correct answer. Write the card number in the 2nd box, and then continue until you reach your first card again.

Student Answer Sheet

Name ____________________

Period _____

Pick a card to start with. Write the number in the 1st box. Work the problem and then locate the card with the correct answer. Write the card number in the 2nd box, and then continue until you reach your first card again.

KEYS

Version A - Set of 10 Cards

1	4	7	2	10	3	8	9	6	5

Note: Students may start with any cards; shift the answer to match the starting point.

Compound Interest Formula

$$\mathcal{A} = P\left(1 + \frac{r}{n}\right)^{nt}$$

Show work below

_1__ $\mathcal{P} = 2.50$ $r = .09$ $n = 1$ $t = 20$ $\mathcal{A} = 2.50(1 + \frac{.09}{1})^{20}$ = \$14.01	_4___ $\mathcal{P} = 500$ $r = .05$ $n = 4$ $t = 5$ $\mathcal{A} = 500(1 + \frac{.05}{4})^{20}$ = \$641.02	_7___ $\mathcal{P} = 200$ $r = .03$ $n = 12$ $t = 10$ $\mathcal{A} = 200(1 + \frac{.03}{12})^{120}$ = \$269.87	__2__ $\mathcal{P} = 1.00$ $r = .0125$ $n = 1$ $t = 27$ $\mathcal{A} = 1(1 + \frac{.0125}{1})^{27}$ = \$1.40
_10___ $\mathcal{P} = 300$ $r = .038$ $n = 12$ $t = 1$ $\mathcal{A} = 300(1 + \frac{.038}{12})^{12}$ = 311.60-300=11.60	_3___ $\mathcal{P} = 50{,}000$ $r = .05$ $n = 1$ $t = 12$ $\mathcal{A} = 50000(1 + \frac{.05}{1})^{12}$ = \$89,793	_8__ $\mathcal{P} = 275$ $r = .029$ $n = 2$ $t = 4$ $\mathcal{A} = 275(1 + \frac{.029}{2})^{8}$ = \$308.57-275=\$33.57	__9__ $\mathcal{P} = 120{,}000$ $r = .065$ $n = 1$ $t = $ 2012-2001= 11 $\mathcal{A} = 120000(1 + \frac{.065}{1})^{11}$ = \$239,898
_6___ $\mathcal{P} = 250$ $r = .085$ $n = 4$ $t = $ 65-18=47 $\mathcal{A} = 250(1 + \frac{.085}{4})^{188}$ = \$13,024.74	__5__ $\mathcal{P} = 500$ $r = .06$ $n = 25$ $t = 4$ $\mathcal{A} = 500(1 + \frac{.06}{4})^{100}$ = \$2,216.02		

Key Version B: Set of 5 Cards

1	4	3	2	5

Compound Interest Formula

$$\mathcal{A} = P\left(1 + \frac{r}{n}\right)^{nt}$$

Show work below

__1__	__4__	___3_	__2__	_5___
$\mathcal{P} = 500$	$\mathcal{P} = 200$	$\mathcal{P} = 2.50$	$\mathcal{P} = 250$	$\mathcal{P} = 300$
$r = .06$	$r = .03$	$r = .09$	$r = .085$	$r = .038$
$n = 25$	$n = 12$	$n = 1$	$n = 4$	$n = 12$
$t = 4$	$t = 10$	$t = 20$	$t = 65\text{-}18 = 47$	$t = 1$
$\mathcal{A} = 500(1 + \frac{.06}{4})^{100}$ $= \$2,216.02$	$\mathcal{A} = 200(1 + \frac{.03}{12})^{120}$ $= \$269.87$	$\mathcal{A} = 2.50(1 + \frac{.09}{1})^{20}$ = $14.01	$\mathcal{A} = 250(1 + \frac{.085}{4})^{188}$ $= \$13,024.74$	$\mathcal{A} = 300(1 + \frac{.038}{12})^{12}$ Interest=311.60-300 . =11.60

Annually n=1

Semiannual n=2

Quarterly n=4

Bimonthly n=6

Monthly n=12

Weekly n= 52

Daily n=365

Compounding Interest Formula

Amount in Account

Interest Rate
(as a decimal)

Time
(in years)

$$A = P\left(1+\frac{r}{n}\right)^{nt}$$

Principal
(amount you start with)

Number of times compounded per year

Credits Page:

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